

Port Security Implementation Using GNS3 & kali

Mac Filtering Using Forti & Kali

1. Introduction

Due to the increasing risk of network access attacks such as MAC Spoofing, it has become essential to implement Layer 2 security mechanisms on network switches. Port Security is one of the most effective methods to prevent unauthorized devices from accessing the network by limiting and controlling MAC addresses per port.

In this project, a simulated enterprise network was built using GNS3, containing multiple VLANs for different departments, a FortiGate firewall for Layer 3 security, and end-user devices. The main objective was to implement Port Security on access ports to prevent unauthorized hosts from accessing the network.

2. Project Topology

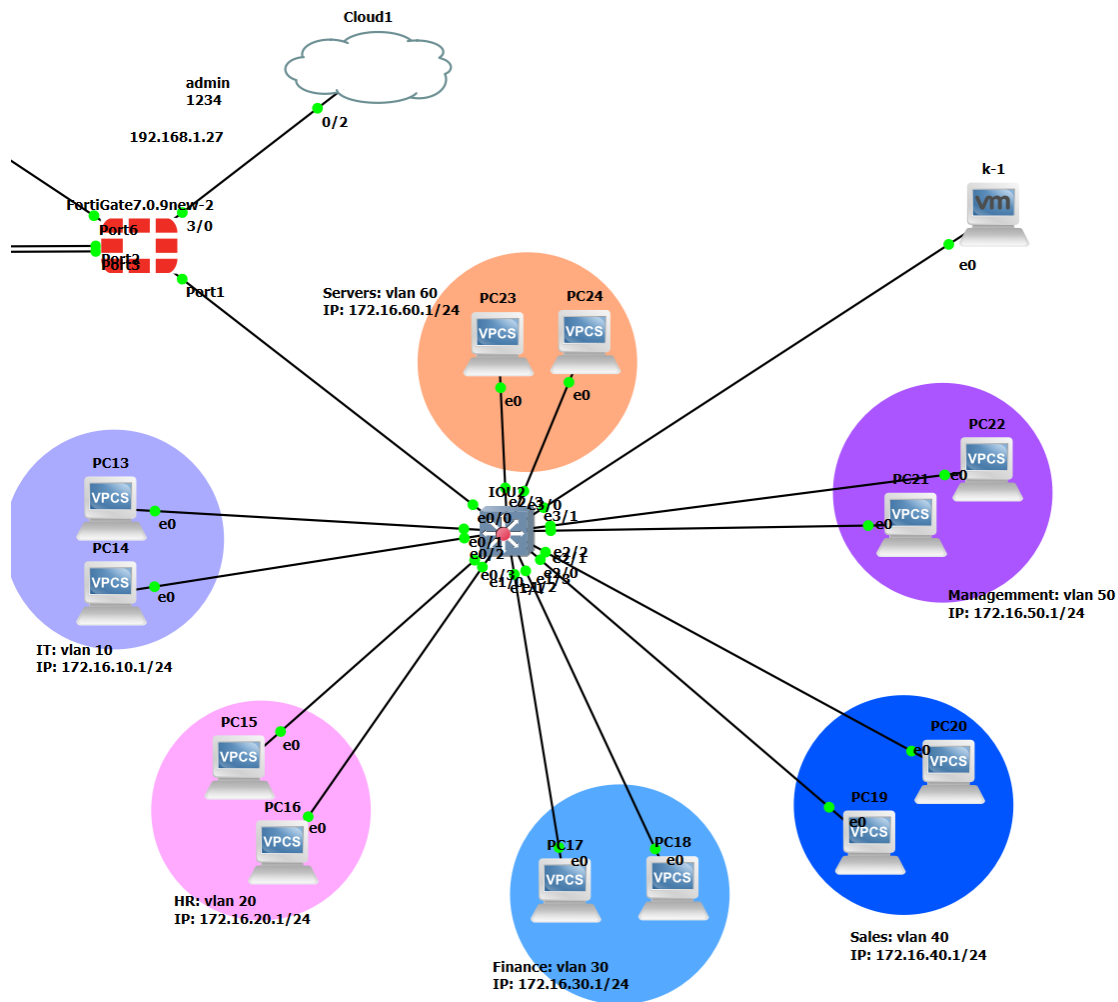
Components Used:

- GNS3 Simulator
- Layer 2 Cisco Switches (IOU/IOS)
- FortiGate Firewall
- End-user PCs distributed on multiple VLANs
- Kali Linux machine to simulate attacker

VLAN Structure:

VLAN	Department	Network
10	IT	172.16.10.0/24
20	HR	172.16.20.0/24
30	Finance	172.16.30.0/24
40	Sales	172.16.40.0/24
50	Manage	172.16.50.0/24
60	Servers	172.16.60.0/24

Each VLAN has dedicated hosts connected through different switch access ports.



3. Problem Statement

A malicious user attempts to gain unauthorized access to the network by spoofing the MAC address of a legitimate host. This allows the attacker to bypass traditional IP-based security and communicate as if they were a trusted device.

Without Port Security, the attacker can:

- Change their MAC address to match a legitimate device
- Join the VLAN
- Communicate with the default gateway
- Access internal resources

To prevent this, Port Security must be implemented on access ports.

4. Attack Scenario (MAC Spoofing Attack)

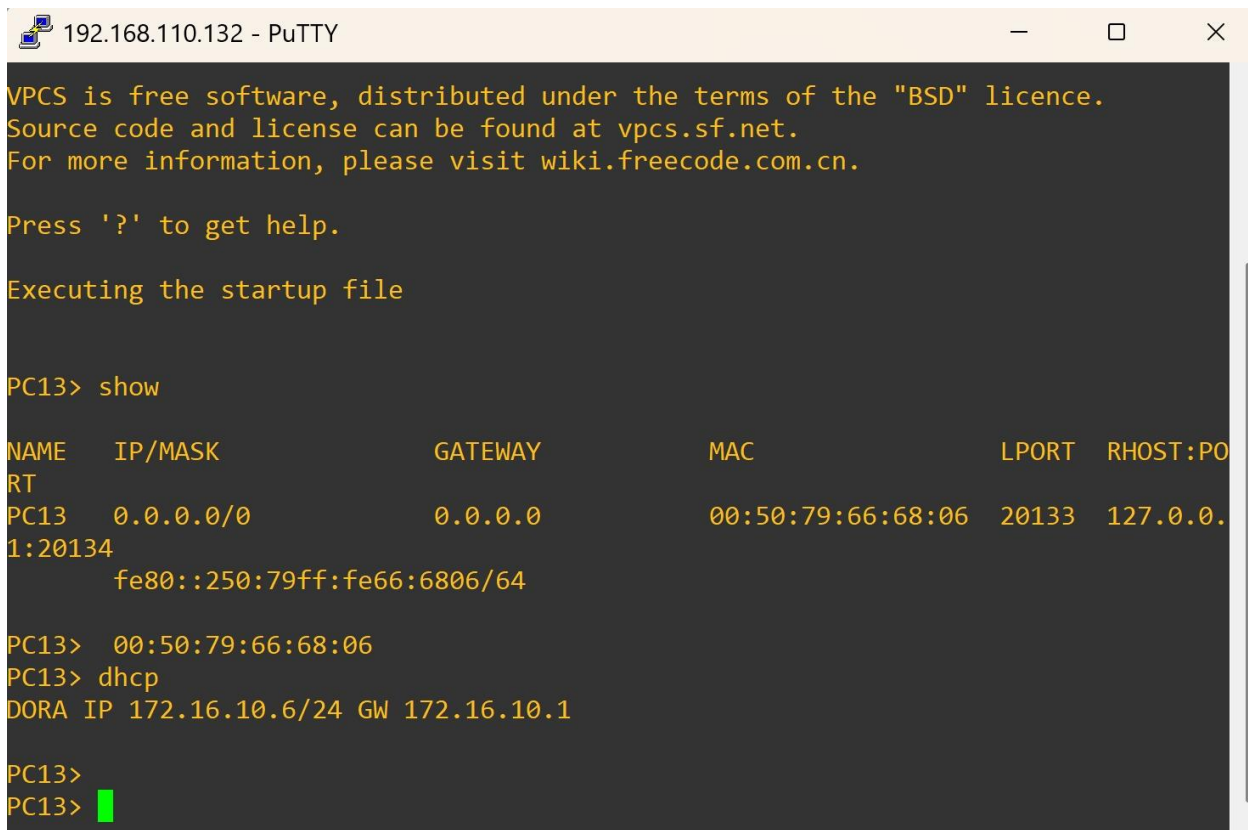
The attacker (Kali Linux) connects to the network and changes their MAC address to match that of a legitimate PC.

Steps performed by the attacker:

```
ip link show eth0
sudo ip link set dev eth0 down
sudo ip link set dev eth0 address < MAC Address>
sudo ip link set dev eth0 up
sudo ip addr add 172.16.10.50/24 dev eth0
sudo ip route add default via 172.16.10.1
ping 172.16.10.1
```

Result before Port Security:

The ping request succeeds → network access is granted → the attack is successful.



The screenshot shows a PuTTY terminal window titled "192.168.110.132 - PuTTY". The terminal output is as follows:

```
VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC13> show

NAME      IP/MASK      GATEWAY      MAC      LPORT  RHOST:PO
RT
PC13      0.0.0.0/0      0.0.0.0      00:50:79:66:68:06  20133  127.0.0.
1:20134
          fe80::250:79ff:fe66:6806/64

PC13> 00:50:79:66:68:06
PC13> dhcp
DORA IP 172.16.10.6/24 GW 172.16.10.1

PC13>
PC13> █
```

```
File Edit View VM Tabs Help
library
Type here to search
My Computer
gns3
Windows Server 2022
gns3
Windows Server 2022
kali@kali: ~
$ ip link show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
link/ether 00:0c:29:97:78:5c brd ff:ff:ff:ff:ff:ff

(kali@kali)-[~]
$ sudo ip link set dev eth0 down
[sudo] password for kali:
(kali@kali)-[~]
$ sudo ip link set dev eth0 address 00:50:79:66:68:06
(kali@kali)-[~]
$ sudo ip link set dev eth0 up

(kali@kali)-[~]
$ ip link show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
link/ether 00:50:79:66:68:06 brd ff:ff:ff:ff:ff:ff permaddr 00:0c:29:97:78:5c

(kali@kali)-[~]
$ sudo ip addr add 172.16.10.50/24 dev eth0
(kali@kali)-[~]
$ sudo ip route add default via 172.16.10.1

(kali@kali)-[~]
$ ping 172.16.10.1
PING 172.16.10.1 (172.16.10.1) 56(84) bytes of data:
64 bytes from 172.16.10.1: icmp_seq=1 ttl=255 time=9.61 ms
64 bytes from 172.16.10.1: icmp_seq=2 ttl=255 time=9.68 ms
64 bytes from 172.16.10.1: icmp_seq=3 ttl=255 time=6.18 ms
64 bytes from 172.16.10.1: icmp_seq=4 ttl=255 time=3.64 ms
64 bytes from 172.16.10.1: icmp_seq=5 ttl=255 time=12.2 ms
64 bytes from 172.16.10.1: icmp_seq=6 ttl=255 time=8.84 ms
64 bytes from 172.16.10.1: icmp_seq=7 ttl=255 time=8.99 ms
64 bytes from 172.16.10.1: icmp_seq=8 ttl=255 time=10.8 ms
64 bytes from 172.16.10.1: icmp_seq=9 ttl=255 time=5.14 ms
64 bytes from 172.16.10.1: icmp_seq=10 ttl=255 time=10.5 ms
64 bytes from 172.16.10.1: icmp_seq=11 ttl=255 time=8.11 ms
^C
  172.16.10.1 ping statistics:
  11 packets transmitted, 11 received, 0% packet loss, time 10018ms
 rtt min/avg/max/mdev = 3.641/8.518/12.182/2.456 ms

(kali@kali)-[~]
$ ss
```

5. Port Security Configuration

Port Security was applied to all access ports connected to end-user devices. This prevents any unauthorized MAC address from connecting to the network.

Configuration Example:

```
enable
conf t
interface e3/1
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
```

end

Explanation:

Command	Function
switchport port-security	enables Port Security
maximum 1	only one MAC allowed
sticky	auto-learns legitimate MAC addresses
violation shutdown	port shuts down if violation occurs

6. Results After Port Security

When the attacker attempted the MAC spoofing attack again, the switch detected a violation and shut down the port automatically.

Output Example:

```
Port Status          : Secure-shutdown
SecurityViolation Count : 1
```

Result:

- The attacker was disconnected
- Network remained protected
- Unauthorized access was blocked successfully

```
IOU2#show port-security int e3/1
Port Security           : Enabled
Port Status             : Secure-shutdown
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 0
Configured MAC Addresses : 0
Sticky MAC Addresses    : 0
Last Source Address:Vlan : 0050.56c0.0002:10
Security Violation Count : 1
```

```
IOU2#
```

```

IOU2(config)#int e2/1
IOU2(config-if)#switchport mode access
IOU2(config-if)#switchport port-security
IOU2(config-if)#switchport port-security maximum 1
IOU2(config-if)#switchport port-security mac-address sticky
IOU2(config-if)#switchport port-security violation shutdown
IOU2(config-if)#do wr
Building configuration...
Compressed configuration from 4013 bytes to 1732 bytes[OK]
IOU2(config-if)#exit
IOU2(config)#int e2/2
IOU2(config-if)#switchport mode access
IOU2(config-if)#switchport port-security
IOU2(config-if)#switchport port-security maximum 1
IOU2(config-if)#switchport port-security mac-address sticky
IOU2(config-if)#switchport port-security violation shutdown
IOU2(config-if)#do wr
Building configuration...
Compressed configuration from 4013 bytes to 1732 bytes[OK]
IOU2(config-if)#exit
IOU2(config)#int e2/3
IOU2(config-if)#switchport mode access
IOU2(config-if)#switchport port-security
IOU2(config-if)#switchport port-security maximum 1
IOU2(config-if)#switchport port-security mac-address sticky
IOU2(config-if)#switchport port-security violation shutdown
IOU2(config-if)#do wr
Building configuration...
Compressed configuration from 4013 bytes to 1732 bytes[OK]
IOU2(config-if)#end
IOU2#
*Dec 5 19:25:21.163: %SYS-5-CONFIG_I: Configured from console by console
IOU2#show port-security
Secure Port  MaxSecureAddr  CurrentAddr  SecurityViolation  Security Action
      (Count)           (Count)           (Count)
-----
Et0/1          1             0             0             Shutdown
Et0/2          1             0             0             Shutdown
Et0/3          1             0             0             Shutdown
Et1/0          1             0             0             Shutdown
Et1/1          1             0             0             Shutdown
Et1/2          1             0             0             Shutdown
Et1/3          1             0             0             Shutdown
Et2/0          1             0             0             Shutdown
Et2/1          1             0             0             Shutdown
Et2/2          1             0             0             Shutdown
Et2/3          1             0             0             Shutdown
Et3/1          1             0             1             Shutdown
-----
Total Addresses in System (excluding one mac per port)    : 0
Max Addresses limit in System (excluding one mac per port) : 4096
IOU2#

```

7. Conclusion

Port Security proved to be an effective method to secure the enterprise network and prevent MAC-based spoofing attacks. By limiting the number of MAC addresses allowed per port and shutting down the port upon violation, unauthorized access was fully eliminated.

This project demonstrates the importance of implementing Layer 2 security alongside firewalls and VLAN segmentation to ensure full network protection.

Objective

Enhance network security on VLAN 30 (Finance Department) by applying **MAC Address Filtering** on the FortiGate firewall to ensure that only **authorized Finance devices** are allowed to access external networks and other VLANs. Any unauthorized MAC addresses attempting to communicate through the firewall are **blocked and logged** for auditing purposes.

2. Network Scope

Component		Details
VLAN Name	Finance	
VLAN ID	30	
IP Subnet	172.16.30.0/24	
Gateway	172.16.30.1	
Firewall Interface	port4	
Security Focus	Restrict Finance VLAN access by MAC Address	

3. Security Requirement

Only authorized Finance PCs should access external networks. Attackers who spoof an IP address inside VLAN 30 must **not** gain access unless their MAC address is explicitly permitted.

This provides **Layer 2 identity validation on top of Layer 3 filtering**, preventing:

- MAC Spoofing
- VLAN hopping
- Lateral movement inside Finance network

4. Configuration Steps

Step 1 – Create Address Object for Authorized MAC

Firewall > Address > Create New > MAC Address

Name : Finance_PC1_MAC

MAC Address : 00:50:79:66:68:0d (example)

This object represents the only approved Finance workstation.

Device (MAC Address) 8					
 Finance_PC1_MAC	00:50:79:66:68:14	 Vlan_30	Address	1	
 HR_PC1_MAC	00:50:79:66:68:0f	 Vlan_20	Address	1	
 HR_PC2_MAC	00:50:79:66:68:02	 Vlan_20	Address	1	
 IT_PC1_MAC	00:50:79:66:68:06	 Vlan_10	Address	1	
 IT_PC2_MAC	00:50:79:66:68:12	 Vlan_10	Address	1	
 Mange_PC1_MAC	00:50:79:66:68:17	 Vlan_50	Address	0	
 SALES_PC1_MAC	00:50:79:66:68:0e	 Vlan_40	Address	0	
 Serv_PC1_MAC	00:50:79:66:68:0c	 Vlan_60	Address	0	

Step 2 – Create Allow Policy for Finance MAC

Policy & Objects > Firewall Policy > Create New

Name : Allow_Finance_By_MAC

Incoming Interface: VLAN_30

Outgoing Interface: port4

Source : Finance_PC1_MAC

Destination : ALL

Service : ALL

Action : ACCEPT

Schedule : always

Enable Log Traffic: Yes

This ensures the legitimate Finance PC is fully authorized.

Step 3 – Create Deny Policy for Unknown MACs

Policy & Objects > Firewall Policy > Create New

Name : Deny_Unknown_Finance_MACs
Incoming Interface: VLAN_30
Outgoing Interface: port4
Source : ALL
Destination : ALL
Service : ALL
Action : DENY
Schedule : always
Log Violation Traffic: Enabled

Important: This policy must be placed *below* the Allow_Finance_By_MAC rule.

This rule blocks any Finance device whose MAC address is not explicitly enrolled.

Vlan_30 → port4 2									
Allow_Finance_By_MAC	Finance_PC1_MAC	all	always	ALL	✓ ACCEPT	✓ Enabled	SSL no-inspection	UTM	
Deny_Unknown_Finance_MACs	Vlan_30 address	all	always	ALL	✗ DENY			✓ All	

5. Testing & Verification

Test Device	MAC	Expected Result
Finance PC	00:50:79:66:68:0d	Allowed – Ping & internet access succeed
Kali Attacker	Random MAC	Denied – Traffic blocked, logged on FortiGate

Kali Commands Used for Spoofing

```
sudo ip link set dev eth0 address 00:50:79:66:68:00
sudo ip addr add 172.16.30.X/24 dev eth0
sudo ip route add default via 172.16.30.1
ping 8.8.8.8
```

Result:

Traffic fails to reach outside VLAN 30 and is logged by FortiGate.

```

(kali㉿kali)-[~]
$ sudo ip addr flush dev eth0
[sudo] password for kali:

(kali㉿kali)-[~]
$ sudo ip addr add 172.16.30.6/24 dev eth0

(kali㉿kali)-[~]
$ sudo ip route add default via 172.16.30.1

(kali㉿kali)-[~]
$ ip link show eth0

2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 00:50:79:66:68:00 brd ff:ff:ff:ff:ff:ff permaddr 00:0c:29:97:78:5c

(kali㉿kali)-[~]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
From 172.16.30.1 icmp_seq=1 Destination Net Unreachable
From 172.16.30.1 icmp_seq=2 Destination Net Unreachable
From 172.16.30.1 icmp_seq=3 Destination Net Unreachable
From 172.16.30.1 icmp_seq=4 Destination Net Unreachable
From 172.16.30.1 icmp_seq=5 Destination Net Unreachable
From 172.16.30.1 icmp_seq=6 Destination Net Unreachable
^C
— 8.8.8.8 ping statistics —
6 packets transmitted, 0 received, +6 errors, 100% packet loss, time 5010ms

```

6. Log Evidence (FortiGate)

You should see entries such as:

Action: DENY
Source MAC: 00:50:79:66:68:00
Interface: VLAN_30
Policy: Deny_Unknown_Finance_MACs
Reason: MAC not authorized

Note on Log Evidence (FortiGate)

The FortiGate **log evidence for denied MAC traffic was not generated**, because **Port Security on the access switch blocked the unauthorized device locally before its traffic could reach the firewall**. As a result, no packets from the attacker were forwarded to the FortiGate, and therefore no deny logs were created at the firewall level.

7. Final Outcome

Security Layer	Result
Port Security	Ensures physical access is restricted per switch port

FortiGate MAC Filtering Adds firewall-level MAC identity verification

Together, both controls provide:

- ✓ Defense-in-depth
- ✓ Protection against rogue devices
- ✓ MAC-level access control beyond switching infrastructure

Conclusion

The Finance VLAN (VLAN 30) is now secured using both **Cisco Port Security** and **FortiGate MAC Filtering**, preventing unauthorized devices from gaining network access even if they spoof valid IP addresses. This solution effectively combines **Layer 2 + Layer 3** security to protect critical financial resources.